

# A Statistical Evaluation of the Various Parameters Governing the Assembly of Solution-Based Nanoparticles on Polymer Substrates

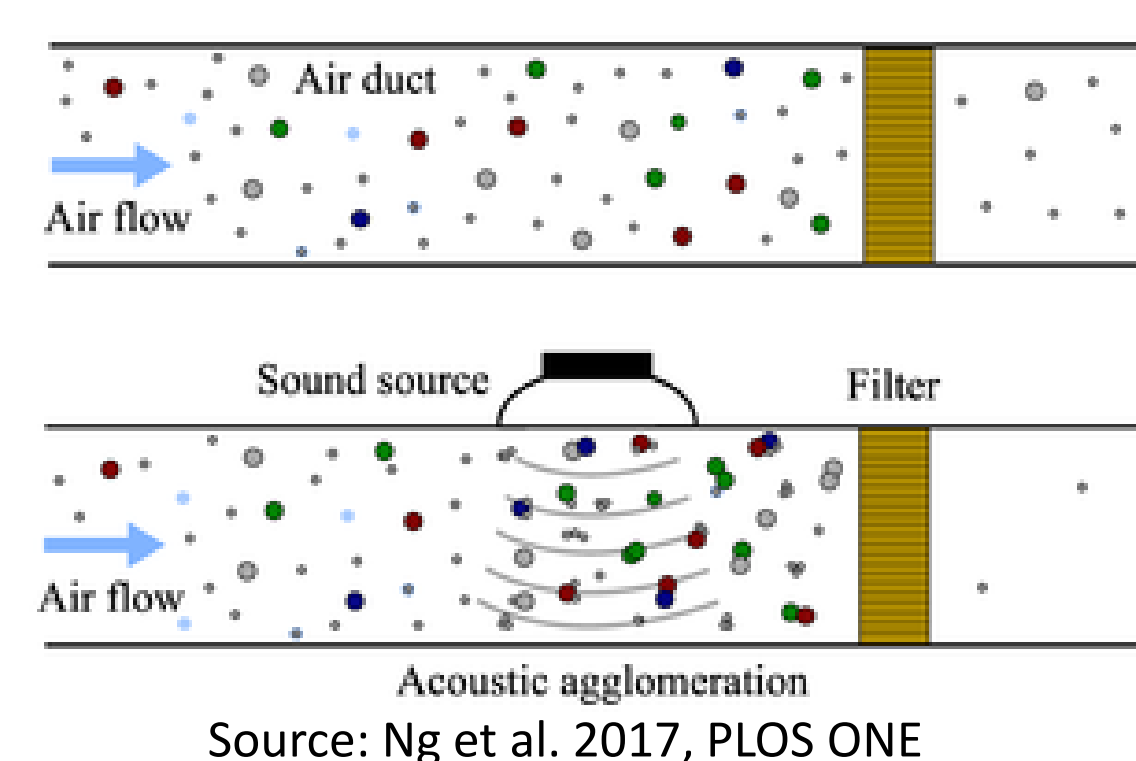
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## Introduction

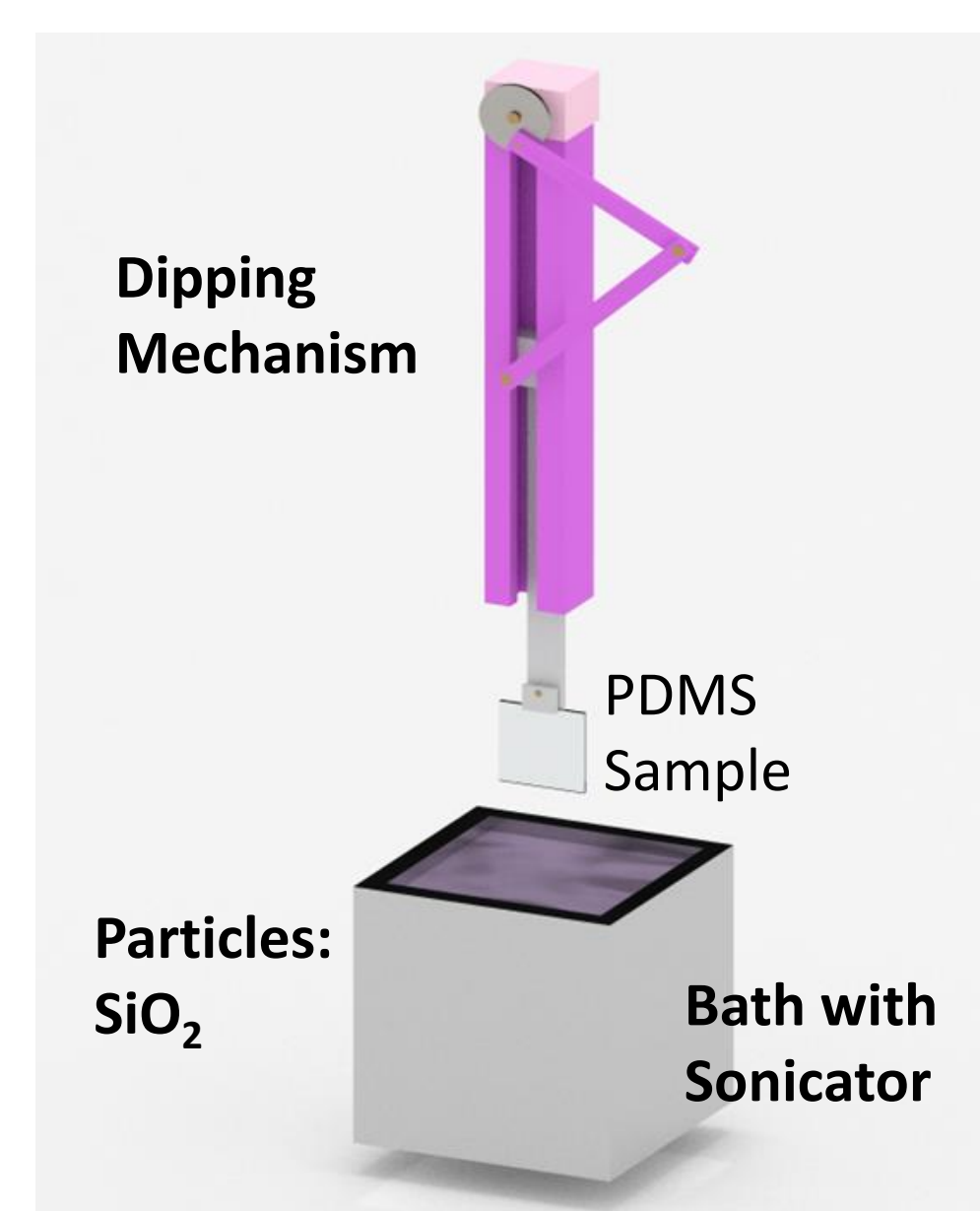
Nanoparticle coating is a breakthrough in material science, enhancing surface properties for use in electronics, photonics, and more. Dip coating, a method for applying nanoparticles, can control thickness to achieve, for example, monolayer nanoparticles. This process, typically involving dip-in and dip-out stages, varies in deposition kinetics due to differing evaporation rates. Understanding these variations is crucial for uniform application. Building on our prior work, we introduced an acoustic field-assisted dip coating technique that ensures uniformity in SiO<sub>2</sub> nanoparticle application without requiring a dip-out stage. We now extend our method to include both dip-in and dip-out processes for thorough evaluation across the entire substrate. Additionally, we use machine learning to predict deposition variability, enhancing control over the nanoparticle assembly process.



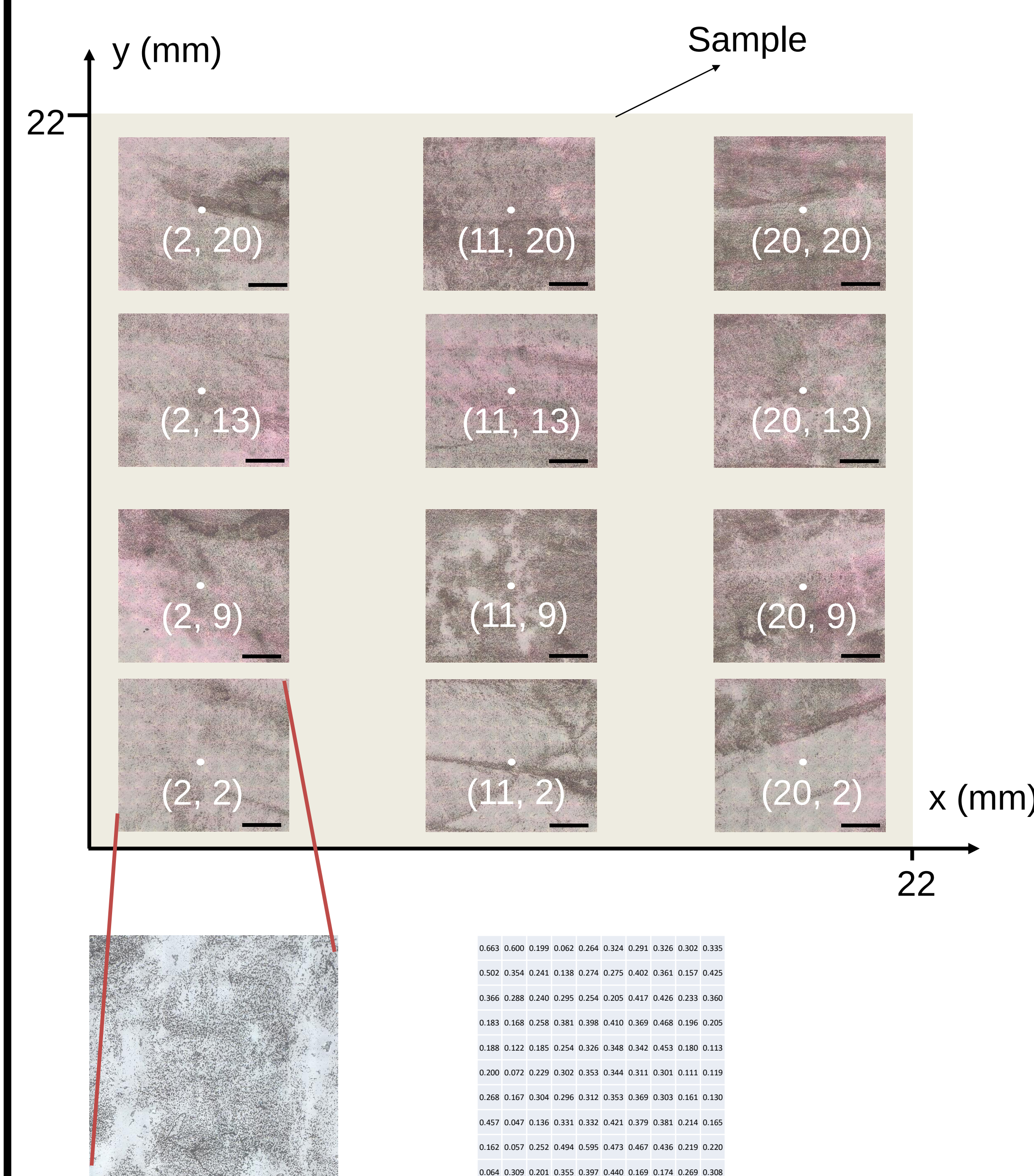
## Experimental Method

Polydimethylsiloxane film (PDMS) is spin-coated on the surface of a glass slide (22 mm x 22 mm). The spin coating speed is brought to 4000 rpm at an acceleration of 500 rpm/s for a duration of 1 min. The ratio of monomer and

crosslinking agent is adjusted to 10:1 in volume. The spin-coated PDMS substrate is then fixed to a holder of a lab-made dip coater. During the dipping process, sonication is applied. Half of the sample is immersed in the solution all the time and the other half experienced the dip-in and dip-out process. The detailed information can be found in the schematic.



## Analyses



In this study, we meticulously analyzed samples subjected to dip coating, employing advanced image processing techniques to assess the coating's uniformity and coverage. The process began with the precise division of the samples into twelve distinct regions, each microscopically photographed to capture detailed characteristics. We carefully recorded the central coordinates and dimensions of each image, laying the groundwork for thorough analysis.

Subsequently, we further segmented each image into

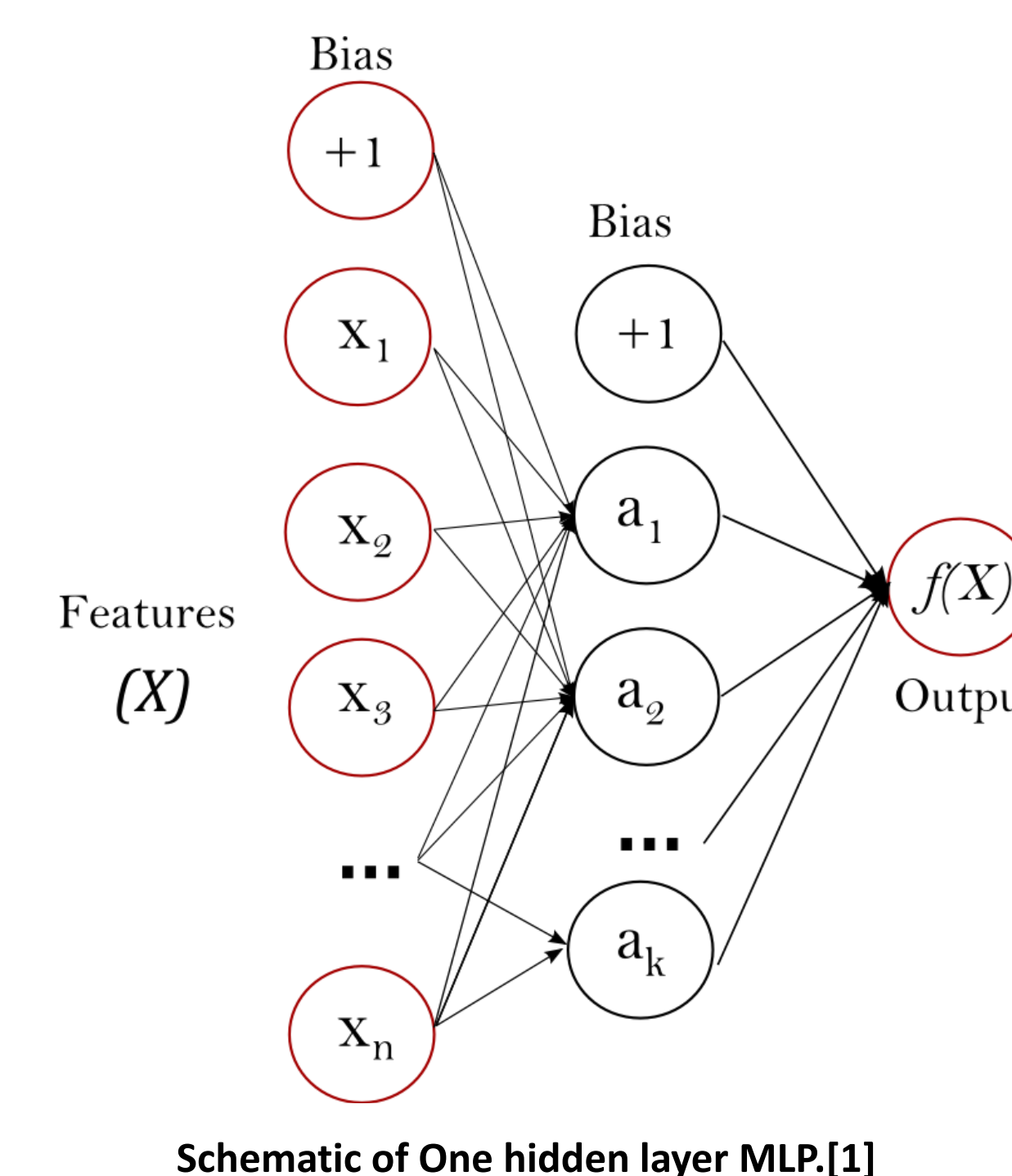
smaller sections, employing image processing methods to quantitatively evaluate the coating coverage across the sample. This detailed approach allowed us to accurately determine the extent of coverage in each area, providing valuable insights into the dip coating process's efficacy and uniformity. This analysis aims to enhance our understanding of coating distribution, contributing significantly to the field's body of knowledge on coating applications.

On the other hand, the focus of the analysis is on the data sets related to Dip coating, which contain a set of features containing the target variables. The feature set consists of "Position x", "Position y", "Time", "Size" and "Concentration", which represent the key intrinsic parameters of the Dip coating process.

Input:

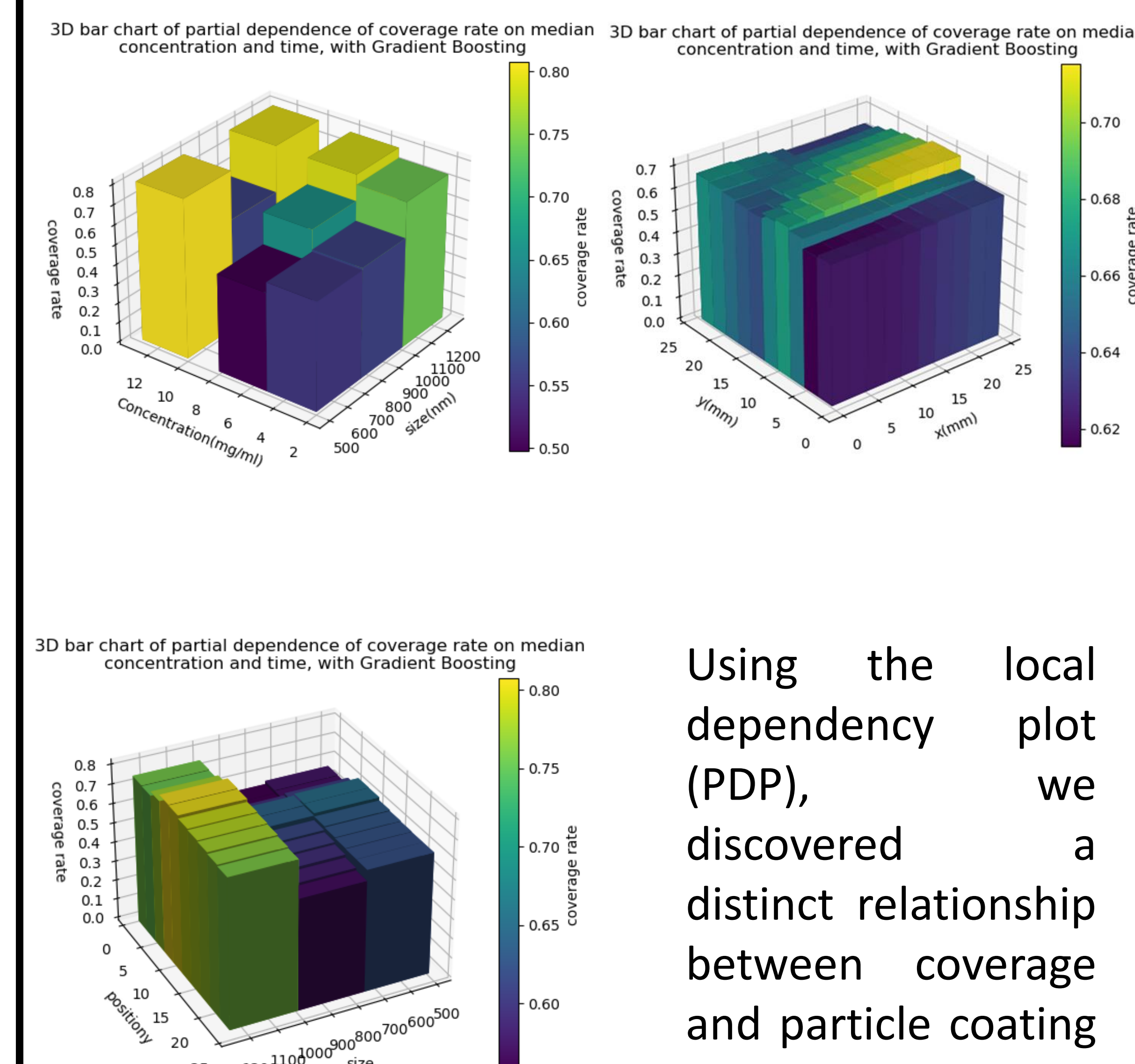
Location  
Time  
Concentration  
Particle size  
\*  
\*

Output:  
Coverage



Schematic of One hidden layer MLP[1]

## Results



Using the local dependency plot (PDP), we discovered a distinct relationship between coverage and particle coating conditions. Notably,

high particle concentrations initially hinder COVERAGE, which then improves as particle SIZE increases. This insight reveals a critical balance in particle concentration and size for optimizing coating uniformity

## References

- Ng BF, et al. Application of acoustic agglomeration to enhance air filtration efficiency in air-conditioning and mechanical ventilation (ACMV) systems. PLOS ONE,(2017)
- \*Zhao, L. et al. Wafer-Scale Full-Coverage Self-Limiting Assembly of Particles on Flexible Substrates. ACS Appl. Mater. Interfaces(2022)

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